Fenchel Dual

primal: $\min: f(x) + g(Ax)$

rewrite: $\min: f(x) + g(y)$

$\text{st}: \quad y = Ax$

dual: $\max_{z}: -f^*(-A^T z) - g^*(z)$

Deriving the dual

$$\boxed{\begin{aligned}&\min: f(x) + g(y)\\ &\text{st}: \; y = Ax\end{aligned}} \longleftrightarrow \min_{x,y}: \max_{z}: f(x) + g(y) + z^T(Ax - y)$$

$$= \max_{z}: \boxed{\min_{x,y}: f(x) + g(y) + z^T A x - z^T y}$$

$$\min_{x}: (A^T z)^T x + f(x) \quad + \quad \min_{y}: (-z)^T y + g(y)$$

$$= -\underbrace{\sup_{x}: (-A^T z)^T x - f(x)}_{f^*(-A^T z)} \quad - \underbrace{\sup_{y}: z^T y - g(y)}_{g^*(z)}$$

$$\boxed{\max_{z}: -f^*(-A^T z) - g^*(z)}$$

Requires strong duality : min max = max min

Slater's : $\exists \tilde{x} \in \text{int dom } f$

$A\tilde{x} \in \text{int dom } g$

Optimality Conditions and Lagrangian

min: $f(x) + g(Ax)$

min: $f(x) + g(y)$

st: $y = Ax$

$\hat{x}$ is an optimal sol'n if

$$0 \in \partial f(\hat{x}) + A^T \partial g(A\hat{x})$$

i.e., $\exists z \in \partial g(A\hat{x})$ with $-A^T z \in \partial f(\hat{x})$

Equivalent to: $\hat{x}, \hat{y} = A\hat{x}$ minimizers of Lagrangian

$$L(x, y, z) = f(x) + g(y) + z^T(Ax - y)$$

$\hat{x}, \hat{y}$ opt. for given z:

$$0 \in \partial f(\hat{x}) + z^T A$$

$$0 \in \partial g(\hat{y}) - z$$

Examples: Set Constraint

$$\min: f(x)$$
$$\text{st}: Ax - b \in C \qquad (x \in \mathcal{X})$$

$$\min: f(x) + \underbrace{I_C(Ax-b)}_{g(Ax)}$$

$$\max: -b^T z - I_C^*(z) - f^*(-A^T z)$$

Sub-Examples:

Ⓐ $\min f(x)$ st: $Ax = b$ | $C = \{0\}$, $I_C^*(z) = 0$

Ⓑ $\min: f(x)$ st: $\|Ax - b\| \leq 1$ | $C = B_{\|\cdot\|}(1)$, $I_C^*(z) = \|z\|_*$

Ⓒ $\min: f(x)$ st: $Ax \preceq_K b$ | $C = -K$, $I_C^*(z) = I_{K^\circ}(z)$

Please check for yourselves

Examples: Norm Regularization

$$\min: f(x) + \|Ax - b\|$$

$$\hookrightarrow \min: f(x) + g(Ax), \qquad g(y) = \|y - b\|$$

$$\max: -f^*(-A^T z) - g^*(z)$$

$$g^*(z) = \, ?$$

Recall: $\left(h(x-b)\right)^*(z) = b^T z + h^*(z)$

$$\left(\|\cdot\|\right)^* = I_{B_{\|\cdot\|_*}(1)}$$

$$= b^T z + I_{B_{\|\cdot\|_*}}(z)$$